

2-Hour Python Lecture Plan

For students with basic Python knowledge | Beginner-friendly & slightly advanced pace

Lecture Overview

Time	Topic
0:00 - 0:10	Quick Recap
0:10 - 0:25	Data Types & Variables (deeper dive)
0:25 - 0:45	Control Flow: if/else & loops
0:45 - 1:05	Functions
1:05 - 1:15	Break
1:15 - 1:35	Lists, Dictionaries & Loops together
1:35 - 1:50	Hands-on Practice
1:50 - 2:00	Wrap-Up & Q&A

1. Quick Recap (10 min)

Ask students to recall:

- What Python is used for (scripting, web, data, automation)
- How to print output: `print("Hello")`
- Basic variable assignment: `x = 5`
- Comments: `# this is a comment`

Tip: Ask questions instead of lecturing — keep it interactive.

2. Data Types & Variables (15 min)

Go a bit deeper than basics — cover the core built-in types:

- `int`, `float` — numbers
- `str` — text, and common string methods (`.upper()`, `.lower()`, `.strip()`)
- `bool` — `True` / `False`
- `type()` function to check a variable's type

Example:

```
name = "Alice"
age = 21
is_student = True
```

```
print(type(name), type(age), type(is_student))
print(name.upper())
```

3. Control Flow: if/else & Loops (20 min)

Cover conditionals first:

```
age = 18
if age >= 18:
    print("You can vote")
else:
    print("Not yet eligible")
```

Then loops — for and while:

```
for i in range(5):
    print(i)

count = 0
while count < 3:
    print("Counting:", count)
    count += 1
```

- range() and how it works
- break and continue (briefly)

4. Functions (20 min)

Explain why we use functions: to avoid repeating code and organize logic.

```
def greet(name):
    return "Hello, " + name + "!"

print(greet("Alice"))
print(greet("Bob"))
```

- def keyword, parameters, return values
- Difference between print() and return
- Default parameter values, e.g. def greet(name="friend")

Break (10 min)

Let students stretch, ask quick one-on-one questions.

5. Lists, Dictionaries & Loops Together (20 min)

Lists — ordered collection of items:

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)

fruits.append("mango")
```

Dictionaries — key/value pairs:

```
student = {"name": "Alice", "age": 21}
print(student["name"])

for key, value in student.items():
    print(key, ":", value)
```

6. Hands-on Practice (15 min)

Give students a small task to build during class:

“Write a program that stores 3 friends' names and ages in a dictionary, then loops through and prints a sentence for each: '<name> is <age> years old.' Use a function to format the sentence.”

Walk around / monitor, help as needed.

7. Wrap-Up + Q&A (10 min)

- Recap key concepts (data types, control flow, functions, lists/dicts)
- Assign light homework: build a simple contact book using a dictionary and functions
- Open floor for questions

Tips for Teaching

- Live-code instead of just showing slides — students learn better watching real-time changes.
- Use `print()` liberally during live demos so students can see what's happening at each step.
- Keep energy up with small interactive tasks every 15-20 minutes rather than one long lecture block.
- Use the break at the 1-hour mark to reset attention for the second half.